4. **Fill in Purely Single-Set Regions:**
    - **Chemistry (C):** Total C = 40. Already accounted for in overlaps: $10 + 18 + 5 = 33$. Purely C = $40 - 33 = 7$.
    - **Physics (P):** Total P = 70. Already accounted for in overlaps: $12 + 18 + 5 = 35$. Purely P = $70 - 35 = 35$.
    - **Mathematics (M):** Total M = 100. Already accounted for in overlaps: $12 + 18 + 10 = 40$. Purely M = $100 - 40 = 60$.

Final Answer

- The number of students who study Mathematics alone is 60.

Therefore, the correct option is C.

Fastest Finger List

This section lists students who answered the quiz question quickly and correctly:

1. Hemachandran
2. B. Dhivyadarshan
3. Xavier Lalmuanawma Hnamte
4. Utkarsh Kumar
5. Jayaramakrishnan Sr
6. Jessica Sharlin Aluvala
7. Advitiya Rao Mastagar
8. Saisricharan Devaraju
9. Shivani Kallappan
10. Aditya M H

Introduction to Relations and Ordered Pairs

Understanding Relations

To understand functions, it's crucial to first grasp the concepts of sets and relations. Relations form the foundational backend for functions.

Ordered Pairs

Every discussion about relations begins with **ordered pairs**. An ordered pair is represented as (X, Y), where the order of elements is significant. This means that (x, y) is generally not the same

Notes

A classic example of an ordered pair is the coordinate of a point in an x, y plane. For instance, the point $(2, 3)$ signifies that to reach this point from the origin, one must travel 2 units along the x-axis and 3 units along the y-axis. This is distinct from $(3, 2)$, which would mean traveling 3 units along the x-axis and 2 units along the y-axis. The order clearly matters in defining the position of a point.

Cartesian Product of Two Sets

Definition of Cartesian Product

The Cartesian product of two sets, say A and B, denoted as $A \times B$, is the set of all possible ordered pairs (x, y) where the first element x comes from set A, and the second element y comes from set B.

Example of Cartesian Product

Let's consider two sets:

- Set A = $1, 2, 3$
- Set B = A, B

To find $A \times B$, we form ordered pairs where the first element is from A and the second is from B: $A \times B$ = $\{(1, A), (1, B), (2, A), (2, B), (3, A), (3, B)\}$

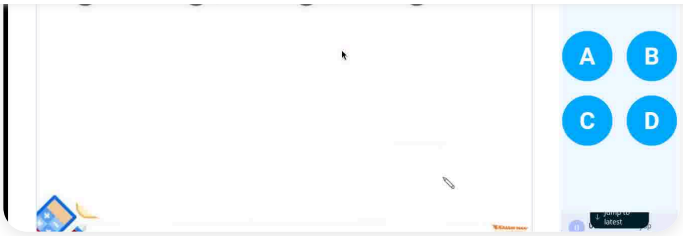
Cartesian Product $B \times A$

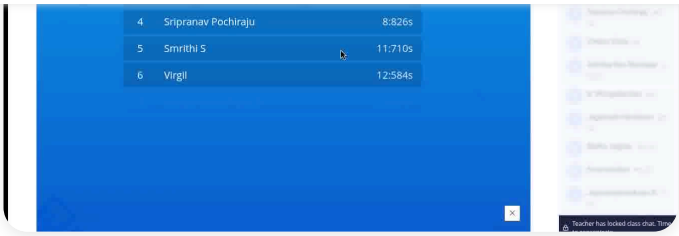
Similarly, for $B \times A$, the first element comes from B and the second from A: $B \times A$ = $\{(A, 1), (A, 2), (A, 3), (B, 1), (B, 2), (B, 3)\}$

Key Properties of Cartesian Product

1. **Order Matters**: In general, $A \times B \neq B \times A$. As seen in the example, the ordered pairs in $A \times B$ are different from those in $B \times A$.
2. **Number of Elements**: The number of elements in the Cartesian product of two sets is the product of the number of elements in each set. $n(A \times B) = n(A) \times n(B)$ Also, $n(A \times B) = n(B \times A)$. In our example: $n(A) = 3$ $n(B) = 2$ $n(A \times B) = 3 \times 2 = 6$ $n(B \times A) = 2 \times 3 = 6$

Quiz: Number of Subsets of a Cartesian Product





Quiz

Q. If $n(A) = 3$ and $n(B) = 2$ then number of subsets of $A \times B$ is A 6 B 2^5 C 2^6 D None

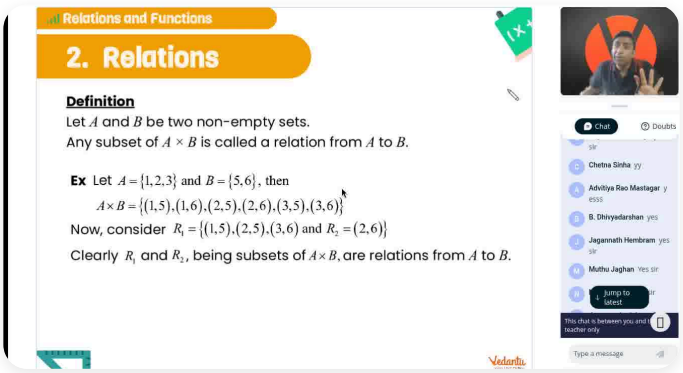
Solution

1. **Find the number of elements in $A \times B$:** Using the formula for the number of elements in a Cartesian product: $n(A \times B) = n(A) \times n(B)$ Given $n(A) = 3$ and $n(B) = 2$. $n(A \times B) = 3 \times 2 = 6$

2. **Find the number of subsets:** The number of subsets of any set with k elements is 2^k. Here, the set is $A \times B$, and it has 6 elements. Therefore, the number of subsets of $A \times B$ is $2^{n(A \times B)} = 2^6$.

Thus, the correct answer is C 2^6.

Definition of a Relation



Understanding Relations

A relation is always defined from two sets, for example, from set A to set B. This is denoted as a relation from A to B.

Let's consider an example:

- Set A = {1, 2, 3}
- Set B = $\{A, B\}$

First, we find the Cartesian product A × B: $A \times B$ = {(1, A), (1, B), (2, A), (2, B), (3, A), (3, B)}

For instance, if we take subsets of A × B:

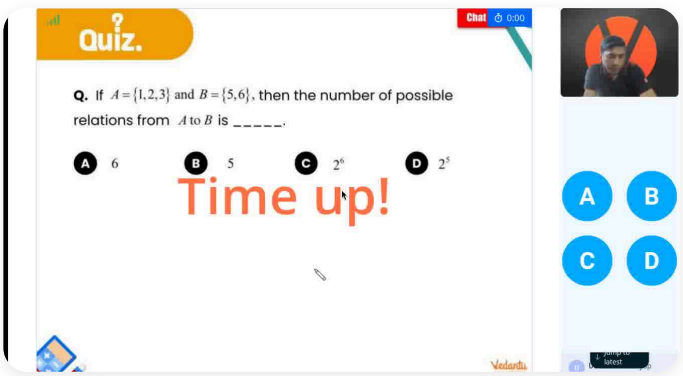
- A null set (empty set) is a subset and thus a relation.
- A subset with exactly one element is a relation.
- A subset with two elements is a relation.
- A subset containing all elements of A × B is also a relation.

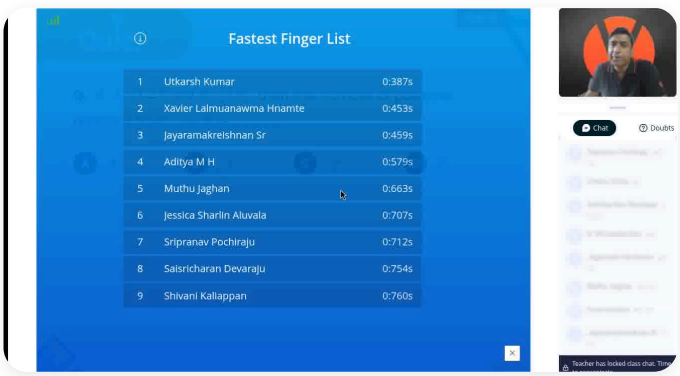
Types of Relations based on Subsets

1. **Empty Relation:** The null set (∅) is an empty relation.
2. **Universal Relation:** The set containing all elements of A × B is a universal relation.

These are important terminologies to be familiar with. Every such subset (R1, R2, R3, R4 in the example) is considered a relation. If a set has a single element, it's called a singleton set, but "singleton relation" is not a standard term.

Total Number of Possible Relations





Calculating the Number of Relations

If a relation is defined from set A to set B, the total number of possible relations is equal to the total number of subsets that can be formed from A × B.

The total number of subsets of any set is given by $2^{\text{number of elements in the set}}$.

Therefore, the total number of relations from A to B is $2^{\text{number of elements in A} \times \text{number of elements in B}}$.

Mathematically, if $n(A)$ is the number of elements in set A and $n(B)$ is the number of elements in set B, then:

$$\text{Total number of relations} = 2^{n(A \times B)} = 2^{n(A) \times n(B)}$$

Quiz Question

Solution

1. **Find the number of elements in set A:** $n(A) = 3$
2. **Find the number of elements in set B:** $n(B) = 2$
3. **Calculate the number of elements in A × B:** $n(A \times B) = n(A) \times n(B) = 3 \times 2 = 6$
4. **Calculate the total number of possible relations:** Total number of relations = $2^{n(A \times B)} = 2^6$

Therefore, the correct answer is **C.** 2^6.

Introduction to Relations

Key Concepts in Relations

In the context of JEE, certain types of relations are particularly important. Direct questions are often asked based on the definitions of these relations. Students are typically given a relation and asked to identify if it is reflexive, symmetric, transitive, or other specific types. This forms a core part of the study of relations.

Describing a Relation

There are three primary ways to describe or represent a relation:

1. **Roster Method (Listing Elements):** This involves explicitly writing down all the ordered pairs that belong to the relation. For example, if a relation R consists of pairs $(1, 2)$, $(3, 4)$, and $(5, 6)$, it would be written as $R = \{(1, 2), (3, 4), (5, 6)\}$.
2. **Set-Builder Notation (Identifying a Pattern):** This method describes the relation by stating a property or a pattern that connects the elements of each ordered pair. For instance, if a relation R from set A to set B has ordered pairs (a, b) such that $b = 2a - 1$, where $a \in A$ and $b \in B$, it can be written as $R = \{(a, b) \mid b = 2a - 1, a \in A, b \in B\}$.
3. **Verbal Description:** This involves stating the relationship between elements in words. For example, "a is related to b such that b is equal to $2a - 1$, where element a belongs to set A."

All three formats are commonly used, and familiarity with each is essential.

Relation as a Subset

In general, any subset of the Cartesian product $A \times B$ is considered a relation from set A to set B. However, in typical problems, a specific condition is provided. This condition helps in identifying a particular subset of $A \times B$, which then defines the relation relevant to the problem. Therefore, while every subset of $A \times B$ is technically a relation, we usually focus on those defined by specific rules or conditions.

Notes

Let's consider a relation R from set A to set B. Suppose set A = {1, 2, 3, 4, 5} and set B = {3, 6, 9, 12, 15}.

If the relation R is given by a set of ordered pairs, for example, R = {(1, 6), (2, 9), (4, 12), (3, 15)}:

- **Domain of a Relation:** The domain of a relation R is the set of all first components (or first elements) of the ordered pairs in R. To find the domain, collect all the first elements from each ordered pair in the relation.
    - For the given relation R, the first components are 1, 2, 4, 3.
    - Therefore, the **Domain of** R = {1, 2, 3, 4}.
    - The domain is also referred to as the **pre-image**.
- **Range of a Relation:** The range of a relation R is the set of all second components (or second elements) of the ordered pairs in R. To find the range, collect all the second elements from each ordered pair in the relation.
    - For the given relation R, the second components are 6, 9, 12, 15.
    - Therefore, the **Range of** R = {6, 9, 12, 15}.
    - The range is also referred to as the **image**.
- **Codomain of a Relation:** The codomain of a relation R from set A to set B is the entire set B. It is the set of all possible second components, regardless of whether they are actually mapped to by an element in the domain.
    - For the given relation R from set A to set B, the **Codomain of** R = B = {3, 6, 9, 12, 15}.

It is important to note that the range is always a subset of the codomain ($Range \subseteq Codomain$). These terminologies are standard and fundamental to understanding relations.

Next Steps and Assignments

Upcoming Session: Relations

- The next session will focus on applications of relations directly relevant to JEE Mains.

Assignment on Sets

- A pinpointed assignment on sets has been provided.
- Students are encouraged to complete it after taking a break and resting.
- The assignment is available on the dashboard and will also be shared on the Telegram channel.

Notes

- Attend sessions with breaks in between to avoid energy depletion.
- Follow the daily dedicated assignments.
- The assignment for Set 1 is available and can be accessed directly or via the Telegram channel.

Session Recap and Acknowledgements

- The instructor confirmed understanding of the session's content with the students.
- A score chart was displayed, acknowledging top performers like Utkarsh, Sai, Jaya, Aditya, Virgil, Jagannath, Diva, Divya, Murthu, Sarvarubali, Adwiti, Xavier, Shivani, Sai, Dhanushin, and Jaswana.
- The instructor thanked all participants for their hard work and participation.